

UNIT-3: Ordinary differential equations of higher orders

CO3: Will able to solve problem involving ordinary differential equations of higher orders.

PART – A (5*2=10)

A1: Solve $D^4y + 16y = 0$.

A2: Write the mathematical structure of Legendre's differential equation.

A3: Define singular point.

A4: find PI $D^2y - 3Dy + 2y = e^{x+1}$.

A5: Why series solution is needed for differential equations.

PART – B (3*7=21)

B1: Solve Cauchy-Euler equation

$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = \log x \sin(\log x).$$

Ans:

$$y = c_1 \cos(\log x) + c_2 \sin(\log x) \\ - \frac{1}{4} (\log x)^2 \cos(\log x) \\ + \frac{1}{4} \log x \sin(\log x).$$

B2: Solve using variation of parameter

$$\frac{d^2y}{dx^2} + y = \operatorname{cosec} x.$$

Ans:

$$y = C.F + .P.I. = c_1 \cos x + c_2 \sin x \\ - x \cos x + x \sin x \log \sin x.$$

B3: Solve in series

$$\frac{d^2y}{dx^2} + x^2y = 0$$

Ans:

$$y = a_0 \left(1 - \frac{1}{12}x^4 + \frac{x^8}{12 \times 7 \times 8} - \frac{x^{12}}{12 \times 8 \times 7 \times 11 \times 12} + \dots \right) + a_1 \left(x - \frac{x^5}{20} + \frac{x^9}{20 \times 8 \times 9} \dots \right)$$

PART – C (3*11=33)

C1: Test for consistency of the following system of equations and if possible, find the solution

$$x + y + z = 8, 2x + 3y + 2z = 19, 4x + 2y + 3z = 23$$

Ans.: 2, 3, 3

C2: Solve the Bessel's equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2)y = 0.$$

C3: Solve in series

$$3x \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y = 0$$

Ans.:

$$y = a_0 \left(1 - \frac{x}{2} + \frac{x^2}{20} - \frac{x^3}{480} + \dots \right) + b_0 x^{1/3} \left(1 - \frac{x}{4} + \frac{x^2}{56} - \frac{x^3}{1680} + \dots \right).$$
